

ChainBrawl

Fully on-chain autobattler where every fight is computed and verified on Solana

The Problem & Our Solution

Most blockchain games fake decentralization: the shop is on-chain, but gameplay runs on centralized servers players can't verify.

- Blockchain games often only put NFTs/items on-chain, not actual gameplay logic
- Centralized battle servers mean players must trust outcomes with no way to verify fairness
- ChainBrawl moves battle resolution itself fully on-chain in an Anchor program
- Fighter stats, combat logic, and randomness all live and execute on Solana
- Result: every fight is deterministic, replayable, and provably fair



How It Works

Mint a fighter, queue for a match, and let an on-chain program resolve combat using VRF randomness, then climb the ladder.

- User flow: mint fighter NFT, stake SOL/tokens, enter match, watch on-chain replay
- Architecture: Anchor program + Metaplex NFT + Switchboard VRF + SPL escrow + React client
- On-chain: fighter stats stored on-chain, combat math executed in-program, VRF seeds randomness
- On-chain: escrow holds stake, leaderboard PDA records rank, payout settled on-chain
- Client is a thin viewer: it reads on-chain state and renders a replay, no hidden server logic

Built with

Solana, Anchor, Switchboard VRF,
Solana Program Library (SPL), Metaplex
NFT, React, Solana Web3.js

Why Now & What's Next

Solana finally makes real-time, verifiable on-chain game logic feasible without ruining UX.

Category

Gaming

- Solana's low fees and fast finality enable per-match on-chain computation at game speed
- Switchboard VRF gives trustworthy, verifiable randomness for fair combat outcomes
- Next: matchmaking and seasonal tournaments with prize pools
- Next: expand to 3v3 multiplayer team battles
- Next: open an SDK so other studios can plug in on-chain battle resolution